

PRESENTATION TOPIC: HTTP VERSION 1

Course Title: Data Communication

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Presented By :

Al-Shahariar

ID: IT-24057

Dept. of ICT,

MBSTU

Supervised By :

Dr. Nazrul Islam

Associate Professor,

Dept. of ICT,

MBSTU

PRESENTATION OUTLINE

01 Introduction: What and Why HTTP/1

02 History of HTTP/1

03 Communication Stack and Working Process

04 Request Methods and Message Format

05 HTTP/0.9 vs 1.0 vs 1.1 Comparison

06 Status Codes, Advantages and Limitations

07 Summary

WHAT IS HTTP/1

- HTTP is the **Hypertext Transfer Protocol** of the World Wide Web.
- It is an **application-layer** protocol that runs over **TCP**.
- It follows a **client–server, request–response** model.
- Messages are **plain text** and **stateless**.
- HTTP/1 covers versions **0.9, 1.0 and 1.1**.

WHY HTTP IS USED

- Fetches **web pages, images, and files** from servers.
- **Simple** and easy to read, write, and debug.
- Works with **any content type** via MIME headers.
- Supported by **every browser, server, and proxy**.
- Foundation of **websites, REST APIs, and web services**.

HISTORY OF HTTP/1

1991	HTTP/0.9 by Tim Berners-Lee at CERN — GET only, HTML only.
1996	HTTP/1.0 (RFC 1945) adds headers, status codes, POST and HEAD.
1997	HTTP/1.1 (RFC 2068) adds persistent connections and Host header.
2022	HTTP/1.1 restated in RFC 9110 and RFC 9112.

HTTP/1 COMMUNICATION STACK

WEB BROWSER (CLIENT)

Chrome, Firefox, curl — sends the request.

HTTP (APPLICATION LAYER)

Request and response text messages.

TCP (TRANSPORT LAYER)

Reliable connection — port 80 by default.

IP (NETWORK LAYER)

Addressing and routing of packets.

WEB SERVER

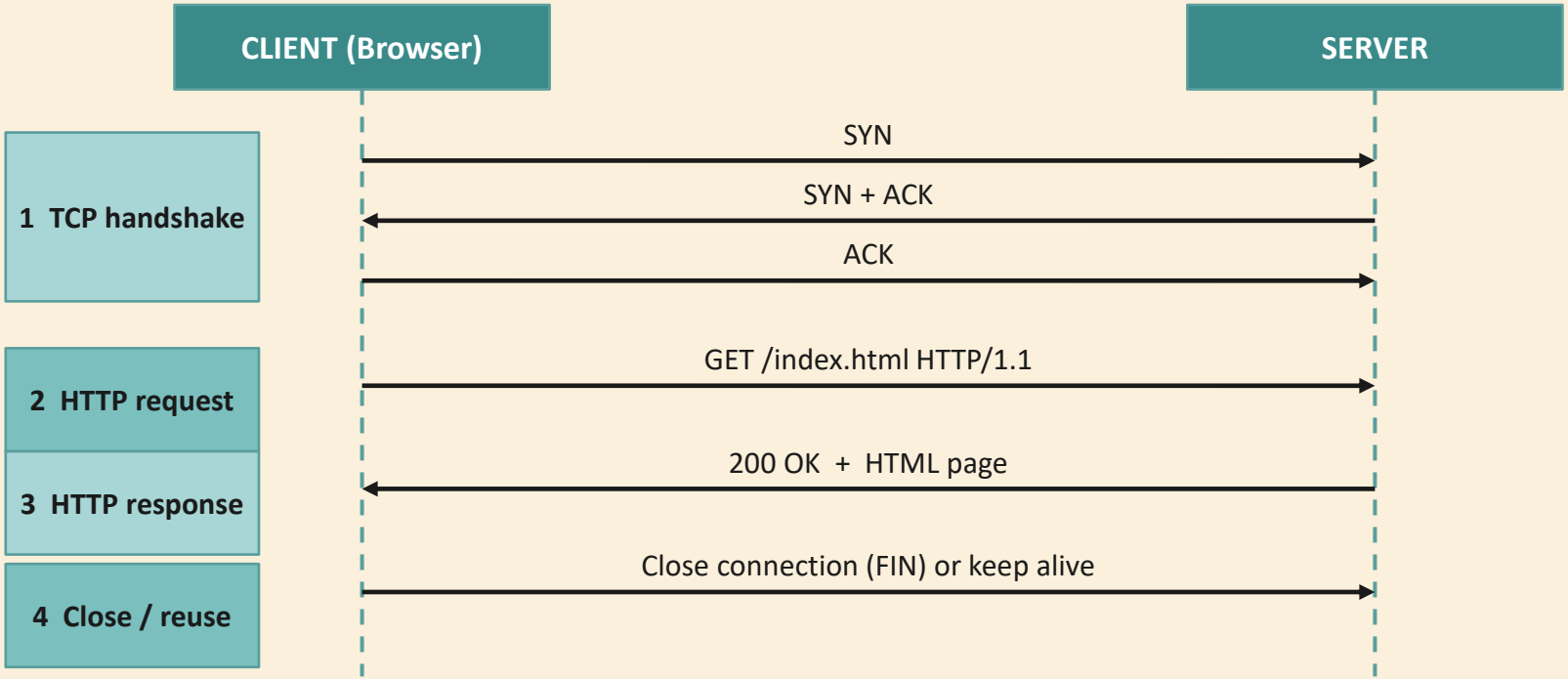
Apache, Nginx, IIS — sends the response.

HOW HTTP/1 WORKS

HTTP Request–Response Cycle — Step by Step:

1. DNS Lookup — Browser resolves the domain to an IP address.
2. TCP Handshake — Client opens a connection to the server.
3. HTTP Request — Client sends method, URL, and headers.
4. HTTP Response — Server returns status code, headers, and body.
5. Close or Reuse — Closed in 1.0; kept alive in 1.1.

HTTP REQUEST–RESPONSE DIAGRAM



HTTP REQUEST METHODS

Actions a client can ask the server to perform:

GET

Use: Retrieve a resource.

POST

Use: Submit data to server.

HEAD

Use: Get headers only.

PUT

Use: Upload or replace (1.1).

DELETE

Use: Remove a resource (1.1).

OPTIONS

Use: List allowed methods (1.1).

HTTP MESSAGE EXAMPLE

HTTP REQUEST

```
GET /index.html HTTP/1.1
Host: www.example.com
User-Agent: Mozilla/5.0
Accept: text/html
```

- Request line (method, URL, version)
- Headers (name: value)
- Blank line ends headers

HTTP RESPONSE

```
HTTP/1.1 200 OK
Content-Type: text/html
Content-Length: 1256

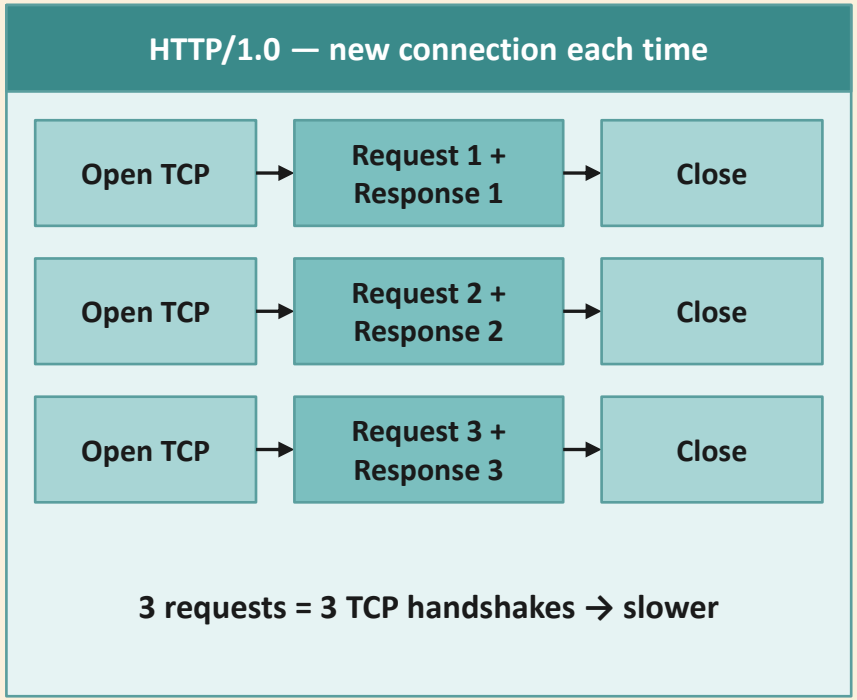
<html> ... </html>
```

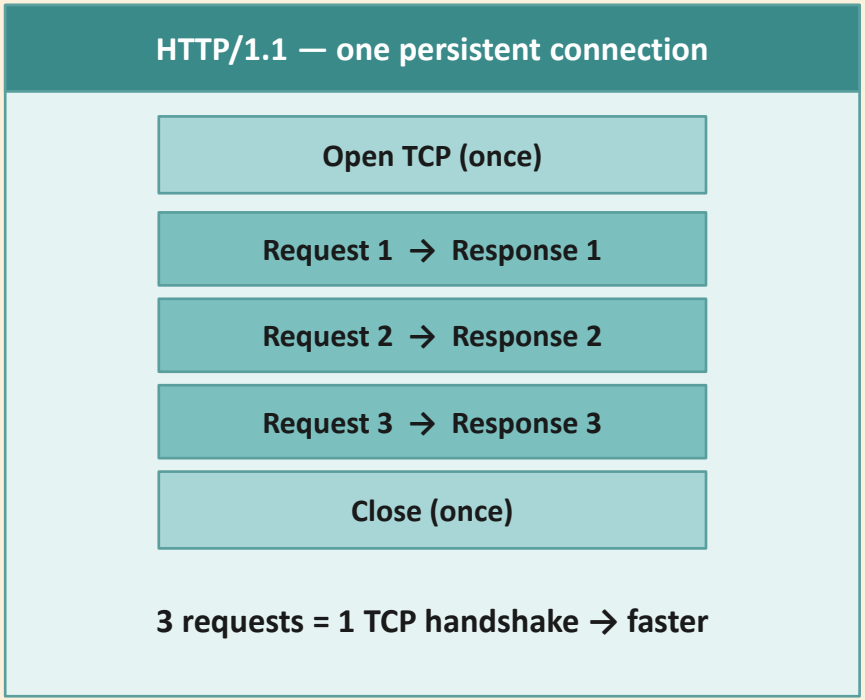
- Status line (version, code, text)
- Headers
- Blank line
- Body (the page content)

HTTP/0.9 vs 1.0 vs 1.1

Feature	HTTP/0.9	HTTP/1.0	HTTP/1.1
Connection	One per request	One per request	Persistent
Methods	GET only	GET, HEAD, POST	Adds PUT, DELETE
Headers	None	Introduced	Host required
Status Codes	None	Introduced	Expanded
Caching	None	Basic	Cache-Control, ETag
Content	HTML only	Any MIME type	Chunked transfer

HTTP/1.0 vs HTTP/1.1 CONNECTIONS





ADVANTAGES OF HTTP/1

- **Simple** and human-readable text format.
- **Persistent connections** reduce delay (1.1).
- **Host header** allows many sites on one IP.
- **Caching** saves bandwidth and time.
- **Universal support** across browsers and servers.

COMMON STATUS CODES

200 OK

Request succeeded; resource returned.

301 Moved Permanently

Resource has a new URL; browser redirects.

304 Not Modified

Cached copy is still valid.

403 Forbidden

Server refuses to allow access.

404 Not Found

Requested resource does not exist.

500 Server Error

Server failed to handle the request.

LIMITATIONS OF HTTP/1

- **Head-of-line blocking** on a connection.
- **Many TCP connections** needed for one page.
- **Repeated, uncompressed** headers waste bandwidth.
- **No encryption** — data is readable by others.
- **Stateless** — needs cookies for sessions.
- **No server push** — later fixed by HTTP/2.

SUMMARY

- HTTP/1 is a **text-based request–response** protocol over **TCP**.
- Versions: **0.9 (1991)**, **1.0 (1996)**, **1.1 (1997)**.
- Core parts: **methods, headers, status codes, and body**.
- HTTP/1.1 added **persistent connections, Host, and caching**.
- Its limits led to **HTTP/2 and HTTP/3**; **HTTPS** adds security.

THANK YOU

Any Questions Are Welcome